THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE UNIT TEST SET A

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. A body of mass 1 kg has weight about ___ on Earth. 1

(a) 1.6 N **(b)** 10 N **(c)** 100 N **(d)** 1 N

2. The weight of an object is the: 1

(a) volume of it **(b)** amount of matter in it

(c) density of it **(d)** force with which Earth pulls it

3. A rubbed plastic scale attracts paper bits due to: 1

(a) electrostatic force **(b)** gravity

(c) friction **(d)** magnetic force

4. Liquids exert pressure: 1

(a) downwards only **(b)** upwards only

(c) in all directions **(d)** sideways only

5. The pressure of a liquid at the bottom increases with: 1

(a) height of the liquid column **(b)** colour of liquid

(c) shape of cap **(d)** weight of container

6. Upper parts of thunderclouds carry mostly: 1

(a) positive charges **(b)** no charges

(c) only neutral dust **(d)** negative charges

7. When a solid is heated, its particles: 1

(a) turn into atoms of another kind **(b)** become bigger

(c) vibrate more vigorously **(d)** stop moving

8. Gas in a syringe can be compressed because: 1

(a) particles are touching **(b)** large interparticle spaces exist

(c) gas has fixed shape **(d)** gas is solid

9. Mishraloha in ancient India referred to a mixture of: 1

(a) two or more metals **(b)** two acids

(c) two gases **(d)** two rocks only

10. A mixture is formed when two or more substances are: 1

(a) mixed without chemical combination **(b)** converted to one element
(c) chemically combined in a fixed ratio **(d)** destroyed

11. The solubility of sugar at 30 °C is 220 g per 100 mL. If 250 g is added to 100 mL, undissolved sugar is: 1

(a) 250 g **(b)** 220 g **(c)** 30 g **(d)** 0 g

12. Solubility of gases in water with rise in temperature: 1

(a) increases **(b)** doubles **(c)** decreases **(d)** stays same

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. What happens when two objects with like charges are brought near? 2

14. Define pressure and write its formula. 2

15. Why does a gas get compressed easily but water does not? 2

16. Define an element with two examples. 2

17. Differentiate solute from solvent. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Why does an object at rest not necessarily have no force acting on it? 3

19. Describe the changes in particle arrangement when ice melts into water and then into vapour. 3

20. Why is rust considered a compound and not a mixture? 3

21. The mass of a stone is 120 g and its volume is 40 cm^3. Calculate density. Will it float in water? 3

22. Explain with an example why a mixture can be separated by physical methods. 3

OR

Is it possible for a substance to be both an element and a compound? Explain. 3

23. Explain the term expansion in the context of water freezing. 3

OR

Explain why the density of a substance decreases when it is heated. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

24. What is atmospheric pressure? Describe two activities that show it. 5

OR

Why do high-speed winds reduce air pressure? Explain with the balloon activity and the roof example. 5

25. Describe an activity to determine the solubility of salt in water at room temperature. 5

OR

Explain with an activity how a raw egg can be made to float in water. 5

END OF QUESTION PAPER Total: 50 Marks